TASK 6

Custom SIEM Security Dashboard

Design and Implementation using the ELK Stack

Internship	Elevance Skills – Cybersecurity Internship
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Technology	Elasticsearch 8.19.0, Logstash 8.19.0, Kibana 8.19.0, Docker Compose
Project type	Educational SIEM dashboard using synthetic security events

Project report

All attack-related events described here are synthetic simulations. No real attacks were performed and no actual data was exfiltrated.

1. Executive Summary

This task designed and implemented a custom Security Information and Event Management (SIEM) dashboard using the ELK Stack. A synthetic dataset of 96 security events was generated, ingested through a dedicated Logstash pipeline, indexed in Elasticsearch, and explored through Kibana. Three Kibana Lens visualizations cover brute-force login attempts, privilege-escalation activity, and potential outbound data exfiltration.

The implementation reused the existing ELK environment from the earlier SIEM correlation-rules task. Task 6 used a separate Logstash container so the existing Task 2 Logstash service did not need to be restarted.

2. Objectives

• Configure a dedicated ingestion pipeline for Task 6 events.

• Generate representative synthetic security events for normal and suspicious activity.

• Index the events in Elasticsearch and expose them through a Kibana data view.

• Create visualizations for brute-force attempts, privilege escalation, and possible exfiltration.

• Combine the visualizations into a centralized SIEM dashboard and validate the results.

3. Tools and Technologies

Elasticsearch 8.19.0 – stores and indexes security-event records.

Logstash 8.19.0 – reads JSON Lines records, parses events, adds a project field, and forwards events.

Kibana 8.19.0 – provides Discover, data views, Lens visualizations, and dashboards.

Docker Compose – runs the ELK containers.

PowerShell – used for dataset generation and administration.

JSON Lines – one JSON event per line for ingestion.

4. Architecture and Data Flow

Flow: Synthetic JSON events → Task 6 Logstash pipeline → Elasticsearch index → Kibana data view → Kibana Lens visualizations → Custom SIEM dashboard.

The Task 6 pipeline is stored at `logstash/task6/task6-dashboard.conf`. It reads `/usr/share/logstash/data-input/task6-dashboard-events.json`, adds `project=task6-custom-siem-dashboard`, and writes to an index matching `siem-dashboard-events-*`. It runs in a separate Logstash container.

5. Synthetic Dataset

Total: **96 events** — 30 normal activity, 26 brute-force simulation, 20 privilege-escalation simulation, and 20 exfiltration simulation.

Fields include `@timestamp`, `event_type`, `scenario`, `event_action`, `event_outcome`, `source_ip`, `username`, `hostname`, `severity`, `process_name`, `command_line`, `destination_ip`, `dns_query`, `bytes_out`, and `description`.

Records are marked as simulations and associated with the lab environment. No real attack traffic was generated.

6. Logstash Ingestion and Elasticsearch

A dedicated Logstash service named `siem-logstash-task6` was configured in Docker Compose. The configuration was validated in test mode with the explicit Task 6 pipeline path; Logstash returned **Configuration OK**.

The pipeline processed the JSON Lines dataset and shut down after reading the file, as configured. Elasticsearch stored the records in an index matching `siem-dashboard-events-*`. An Elasticsearch count query returned **96 indexed documents**.

Kibana data view: Name: SIEM Task 6 Events; index pattern: `siem-dashboard-events-*`; timestamp field: `@timestamp`. Events were visible in Kibana Discover.

7. Dashboard and Visualizations

Dashboard name: Task 6 - Custom SIEM Security Dashboard. The dashboard used the Last 24 hours time range and contained three Lens bar-chart visualizations.

7.1 Brute-force attempts. KQL: `scenario : "brute_force_simulation" and event_outcome : "failure"`. Horizontal axis: `source_ip.keyword`; vertical axis: Count of records. Observed: 25 failed-login events from simulated source IP `203.0.113.45`. The 26-event brute-force simulation also included one successful authentication event.

7.2 Privilege escalation. KQL: `scenario : "privilege_escalation_simulation"`. Horizontal axis: `event_action.keyword`; vertical axis: Count of records. Observed: 12 `process_start` events and 8 `group_membership_change` events, 20 total.

7.3 Potential data exfiltration. KQL: `scenario : "exfiltration_simulation" and event_action : "outbound_transfer"`. Horizontal axis: `destination_ip.keyword`; vertical axis: Sum of `bytes_out`. Observed: approximately 1.23 GB of simulated outbound transfers to `198.51.100.25`.

8. Testing and Validation

Logstash configuration test: Passed (Configuration OK). Elasticsearch connectivity: connected. Dataset ingestion: completed. Indexed event count: 96. Kibana data view: created. Discover visibility: verified. Brute-force, privilege-escalation, and exfiltration visualizations: created. Combined dashboard: created.

9. Limitations and Security Considerations

The records are synthetic and are not real enterprise telemetry. Attack scenarios were simulated; no real brute-force attack, privilege escalation, or data exfiltration was performed.

The dashboard demonstrates filtering, aggregation, and visualization. It does not independently establish whether an event is malicious.

The lab used Elasticsearch with security disabled and is intended for a controlled local environment only. A production deployment would require authenticated access, role-based permissions, secure transport, real log sources, alerting, retention policies, and operational monitoring.

10. Conclusion

Task 6 implemented a custom SIEM dashboard using Elasticsearch, Logstash, Kibana, and Docker. The pipeline ingested 96 synthetic security events, Elasticsearch indexed them, and Kibana displayed them through a dedicated data view and three Lens visualizations. The dashboard provides a consolidated view of simulated brute-force attempts, privilege-escalation activity, and potential outbound data transfers. The task demonstrates the SIEM workflow of event ingestion, indexing, filtering, aggregation, and visualization.

11. Reproducibility Notes

The project includes a PowerShell generator at `scripts/task6/generate-task6-dataset.ps1`, a dataset at `data/task6-dashboard-events.json`, and the Task 6 pipeline at `logstash/task6/task6-dashboard.conf`.

To reproduce ingestion, generate the dataset, validate the pipeline using the explicit configuration path, start the `logstash-task6` Compose service, verify the Elasticsearch count, then use Kibana to create the data view and visualizations.

Repeat-ingestion note: The current file input uses `sincedb_path => "/dev/null"` and reads from the beginning. Running ingestion again can index another copy of the events. Check or remove the prior Task 6 index before repeating ingestion if a clean 96-event count is required.

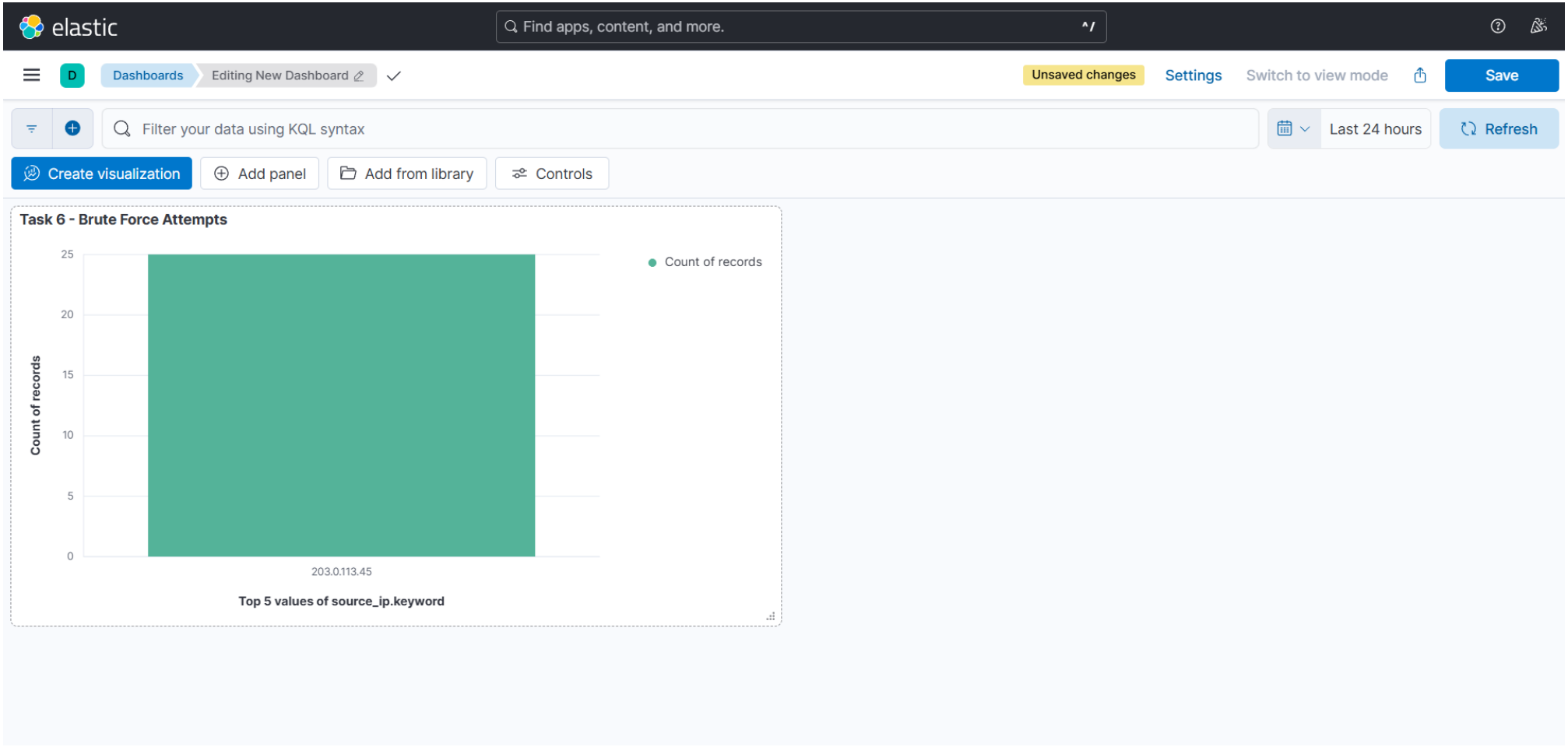
End of report

Appendix A – Implementation Evidence

The following screenshots were supplied as evidence of the Task 6 Kibana implementation and validation.

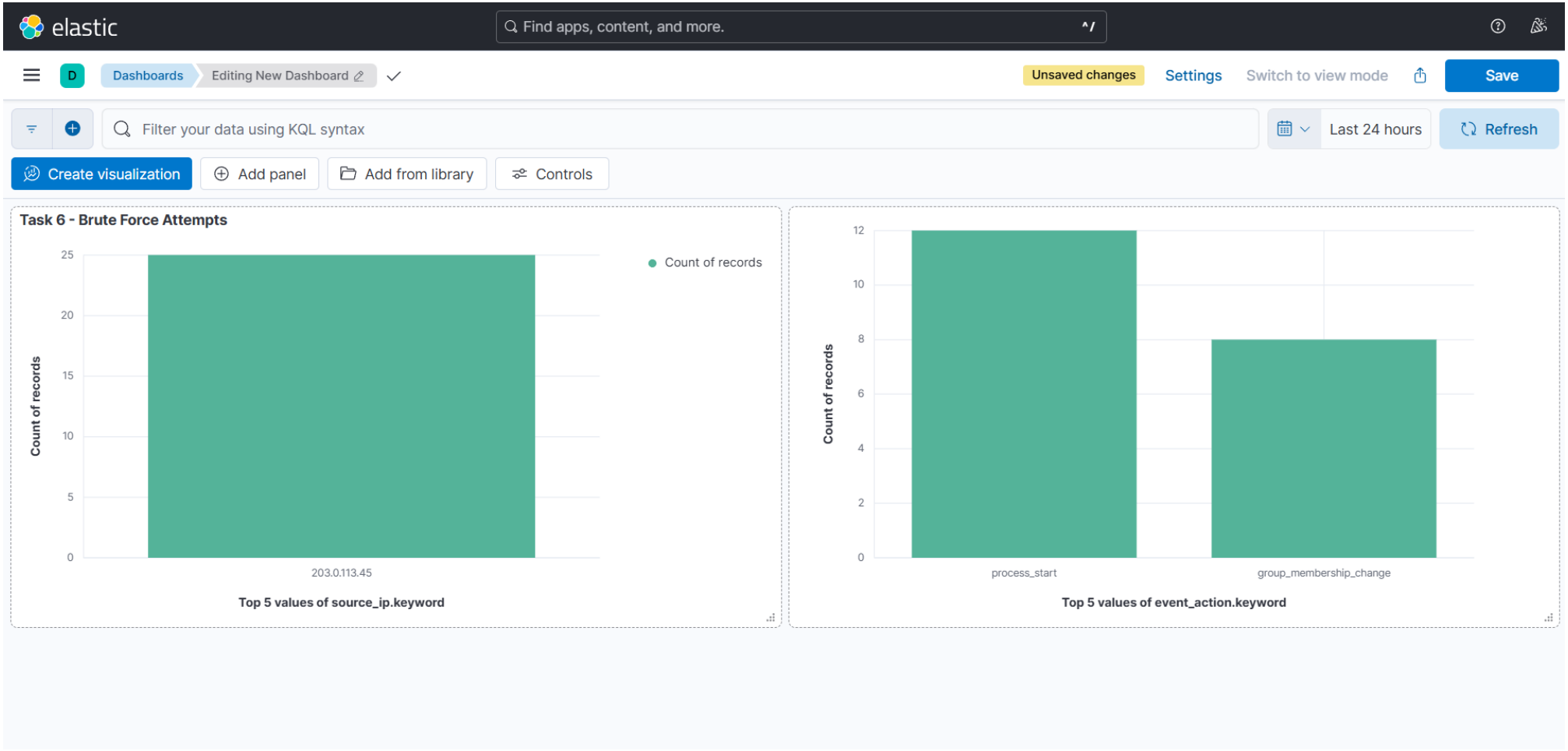
Evidence 1 – Initial brute-force visualization

Kibana dashboard editing view showing the brute-force attempts visualization, with 25 failed events grouped by source IP.



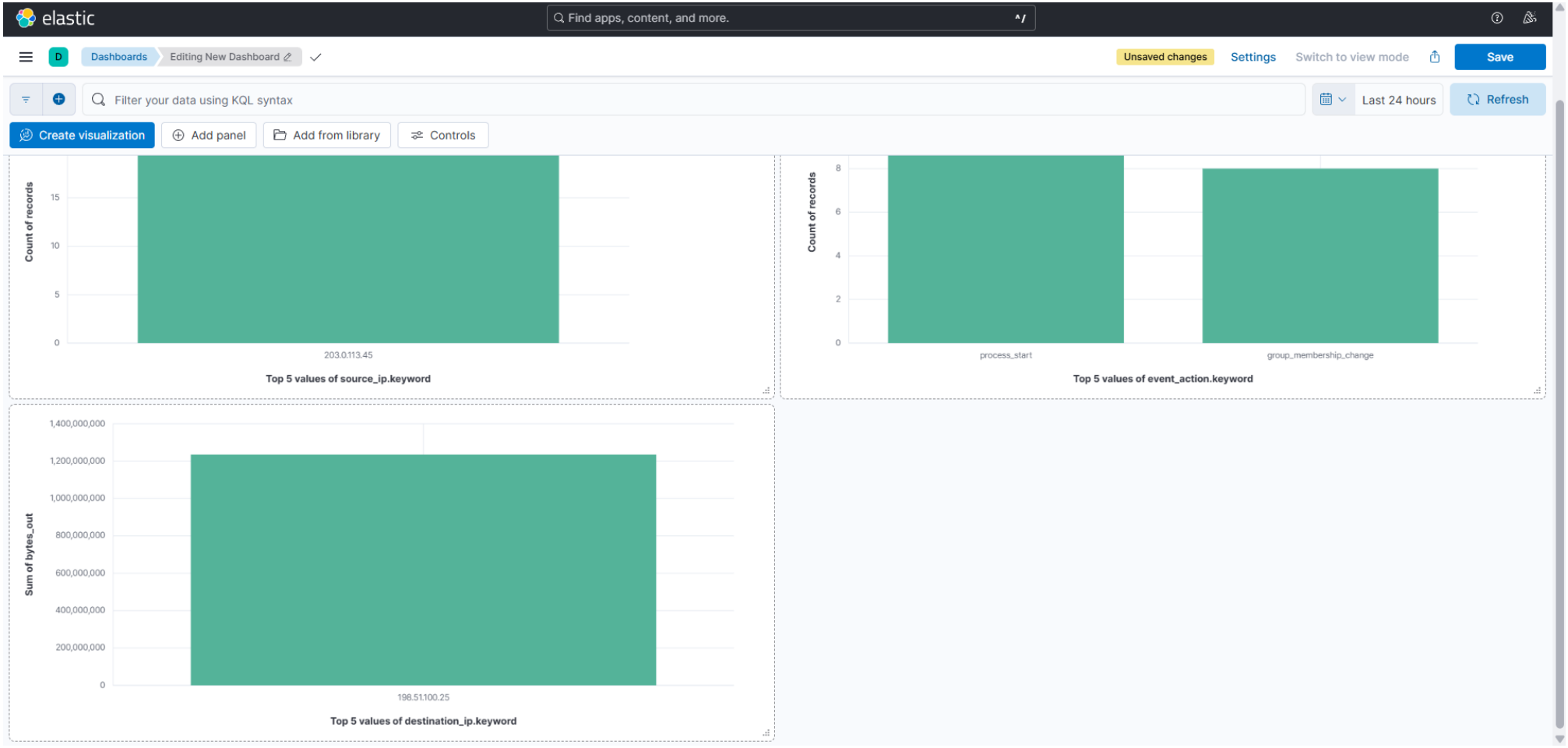
Evidence 2 – Brute-force and privilege-escalation visualizations

Dashboard view showing the brute-force chart and the privilege-escalation chart, including process-start and group-membership-change events.



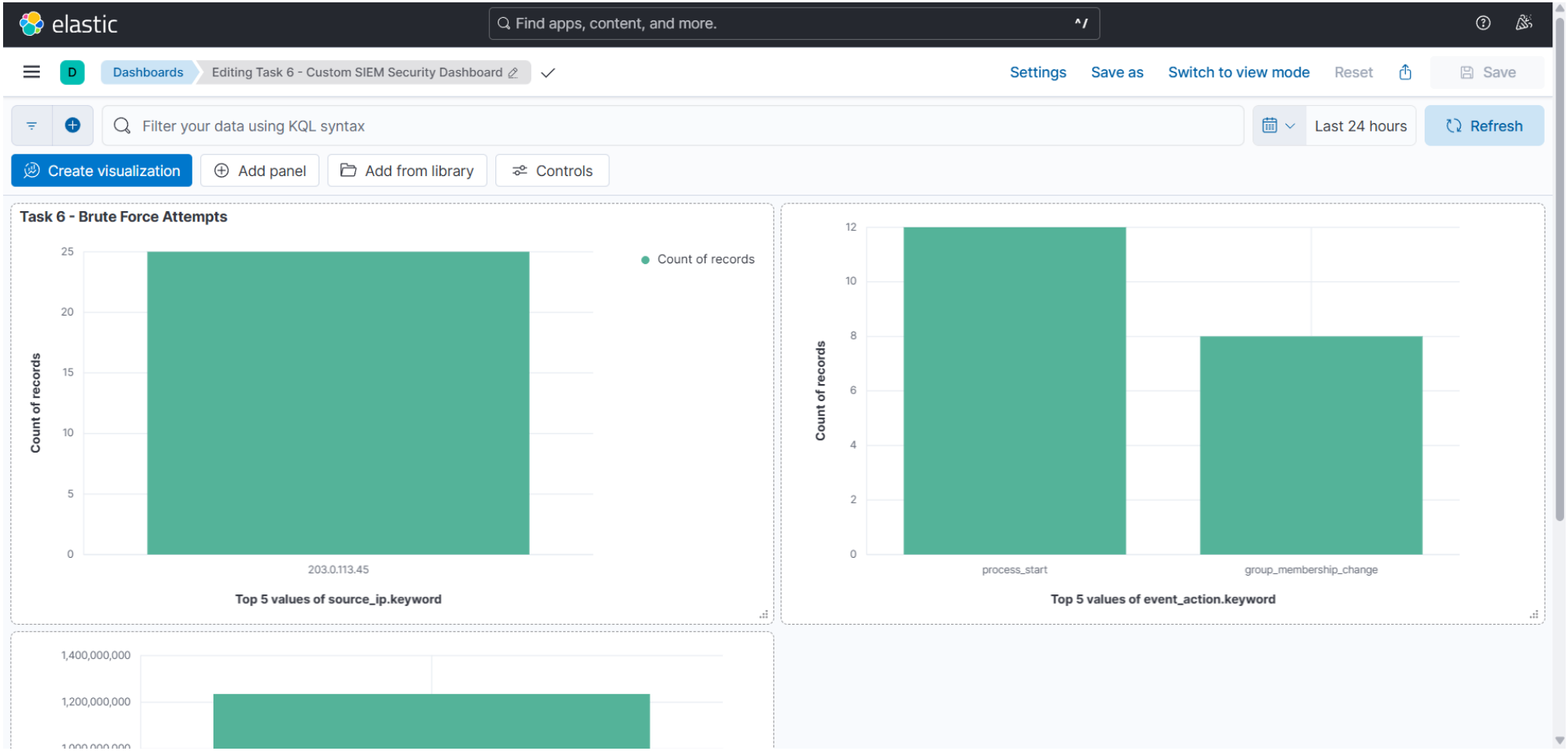
Evidence 3 – Three security-event visualizations

Dashboard view showing brute-force attempts, privilege-escalation activity, and outbound transfer volume grouped by destination IP.



Evidence 4 – Named Task 6 dashboard

Kibana dashboard named “Task 6 - Custom SIEM Security Dashboard,” with the three visualization panels visible.



Evidence 5 – Discover validation

Kibana Discover view using the “SIEM Task 6 Events” data view. The screenshot shows 96 documents.

